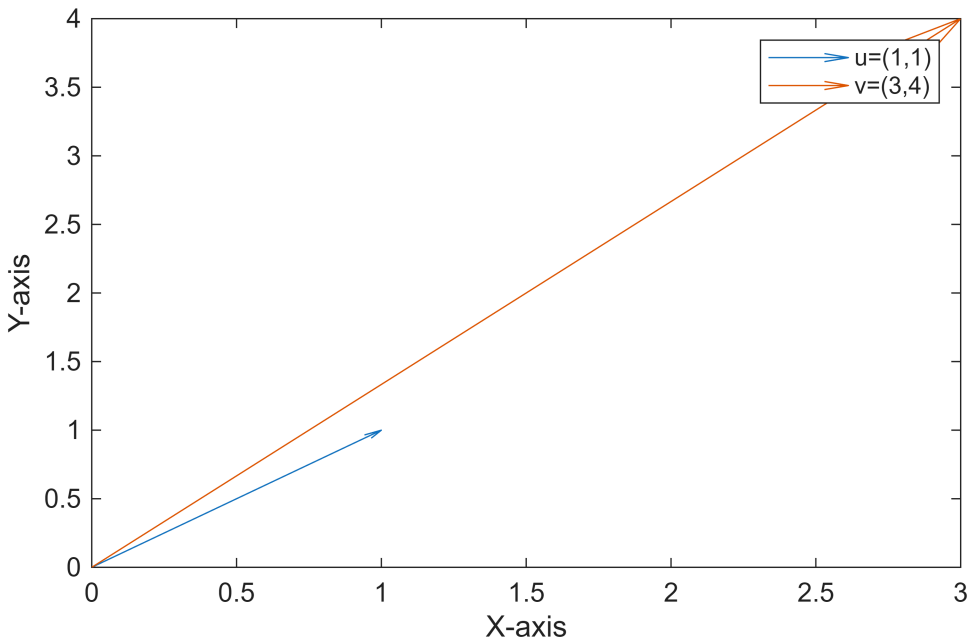
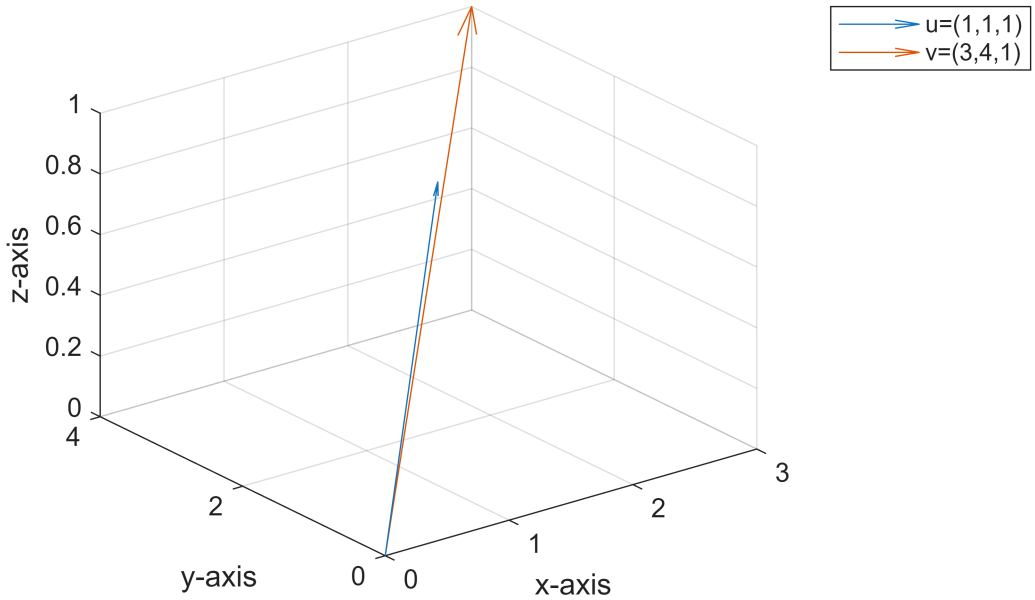
```
%AM.SC.U4CSE25057%
```

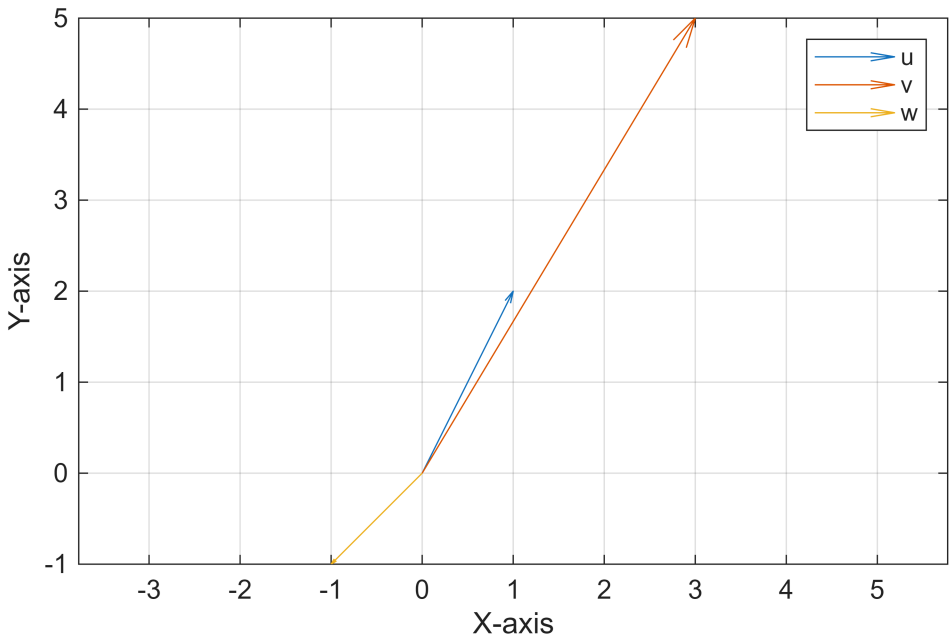
```
quiver(0,0,1,1,0)
hold on
quiver(0,0,3,4,0)
xlabel('X-axis')
ylabel('Y-axis')
legend('u=(1,1)','v=(3,4)')
```



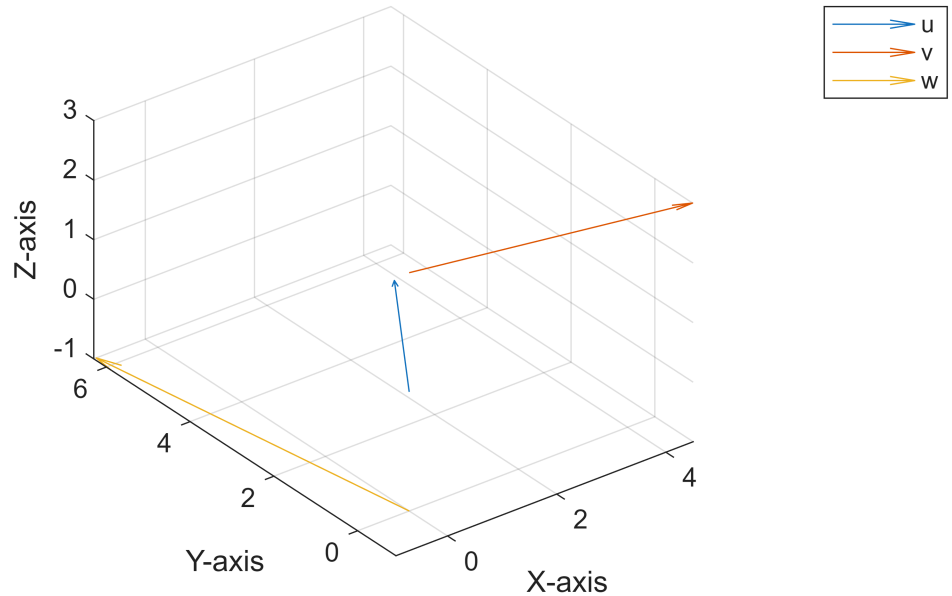

```
clc
clear all
close all
quiver3(0,0,0,1,1,1,0)
hold on
quiver3(0,0,0,3,4,1,0)
view(3)
xlabel('x-axis')
ylabel('y-axis')
zlabel('z-axis')
legend('u=(1,1,1)','v=(3,4,1)')
```



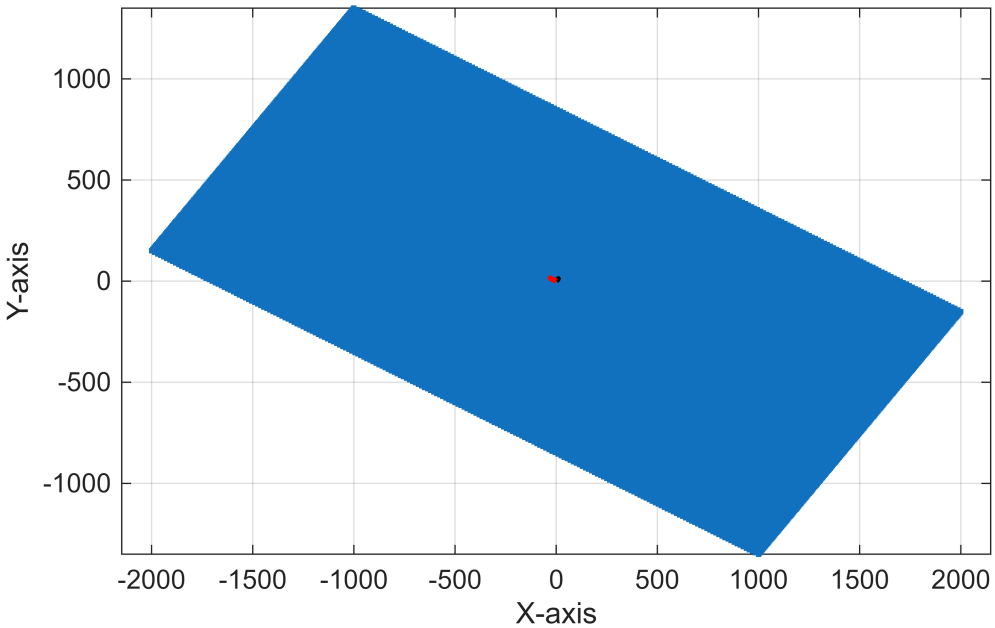

```
clc
clear all
close all
u=[1,2];
v=[3,5];
k1=2;
k2=-1;
w=k1*u+k2*v;
quiver(0,0,u(1),u(2),0)
hold on
quiver(0,0,v(1),v(2),0)
quiver(0,0,w(1),w(2),0)
xlabel('X-axis')
ylabel('Y-axis')
grid on
axis equal
legend('u','v','w')
```



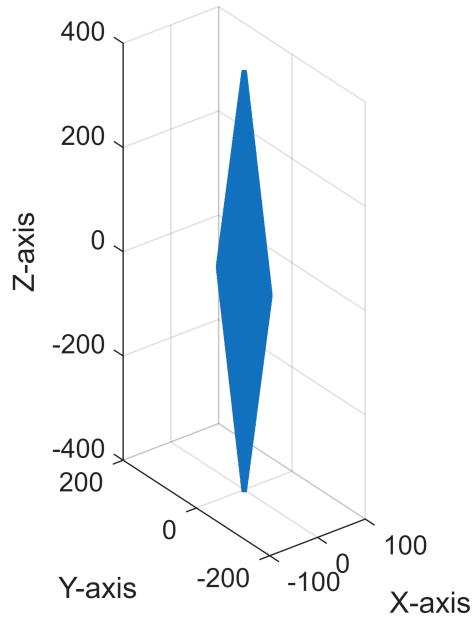

```
clc
clear all
close all
u=[1,2,3];
v=[3,5,-1];
k1=2;
k2=-1;
w=k1*u+k2*v;
quiver3(0,0,u(1),u(2),u(3),0)
hold on
quiver3(0,0,v(1),v(2),v(3),0)
quiver3(0,0,w(1),w(2),w(3),0)
view(3)
xlabel('X-axis')
ylabel('Y-axis')
zlabel('Z-axis')
grid on
axis equal
legend('u','v','w')
```



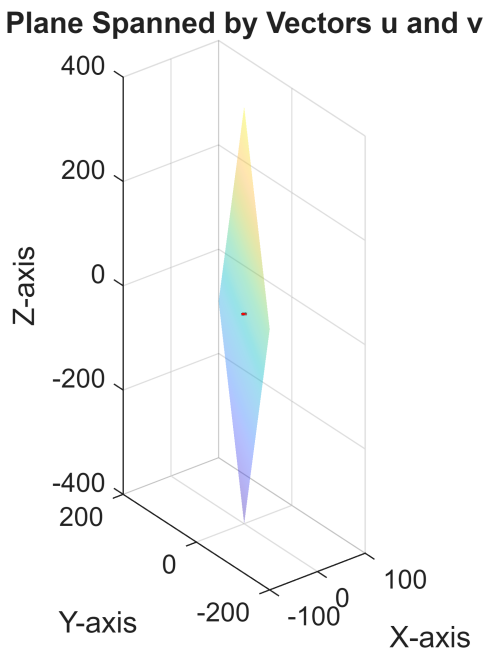

```
%span%
clc
clear all
close all
u=[10,12];
v=[-30,15];
a=-50:0.5:50;
b=-50:0.5:50;
[k1,k2]=meshgrid(a,b);
w=k1(:)*u+k2(:)*v;
plot(w(:,1),w(:,2),'.')
hold on
quiver(0,0,u(1),u(2),0,'k','LineWidth',2)
quiver(0,0,v(1),v(2),0,'r','LineWidth',2)
xlabel('X-axis')
ylabel('Y-axis')
grid on
axis equal
```




```
clc
clear all
close all
u=[1,2,3];
v=[-1,-2,5];
a=-50:0.5:50;
b=-50:0.5:50;
[k1,k2]=meshgrid(a,b);
w=k1(:)*u+k2(:)*v;
plot3(w(:,1),w(:,2),w(:,3),'.')
hold on
quiver3(0,0,u(1),u(2),u(3),0,'k','LineWidth',2)
quiver3(0,0,v(1),v(2),v(3),0,'r','LineWidth',2)
xlabel('X-axis')
ylabel('Y-axis')
zlabel('Z-axis')
view(3)
grid on
axis equal
```

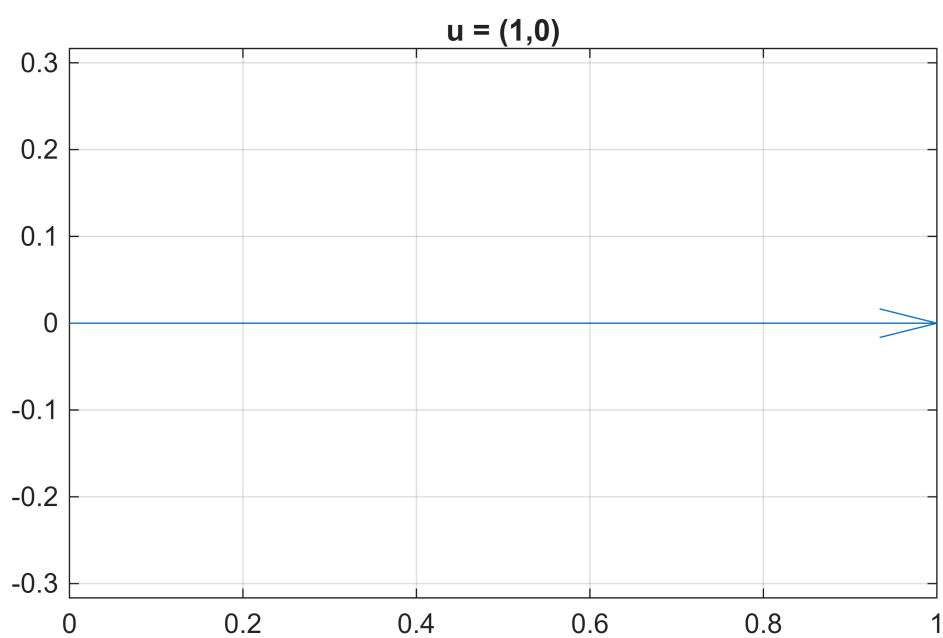



```
clc
clear
close all
u = [1 2 3];
v = [-1 -2 5];
a = -50:0.5:50;
b = -50:0.5:50;
[k1,k2] = meshgrid(a,b);
x = k1*u(1) + k2*v(1);
y = k1*u(2) + k2*v(2);
z = k1*u(3) + k2*v(3);
figure
surf(x,y,z,'FaceAlpha',0.4,'EdgeColor','none')
hold on
quiver3(0,0,0,u(1),u(2),u(3),0,'k','LineWidth',2)
quiver3(0,0,0,v(1),v(2),v(3),0,'r','LineWidth',2)
xlabel('X-axis')
ylabel('Y-axis')
zlabel('Z-axis')
title('Plane Spanned by Vectors u and v')
grid on
axis equal
view(3)
```

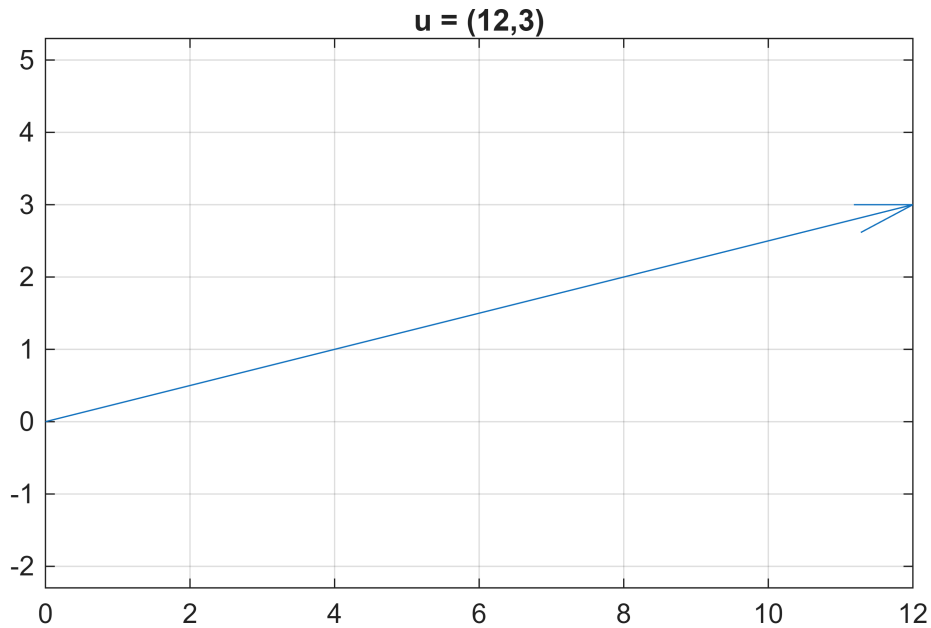



```
%%Question ans
```

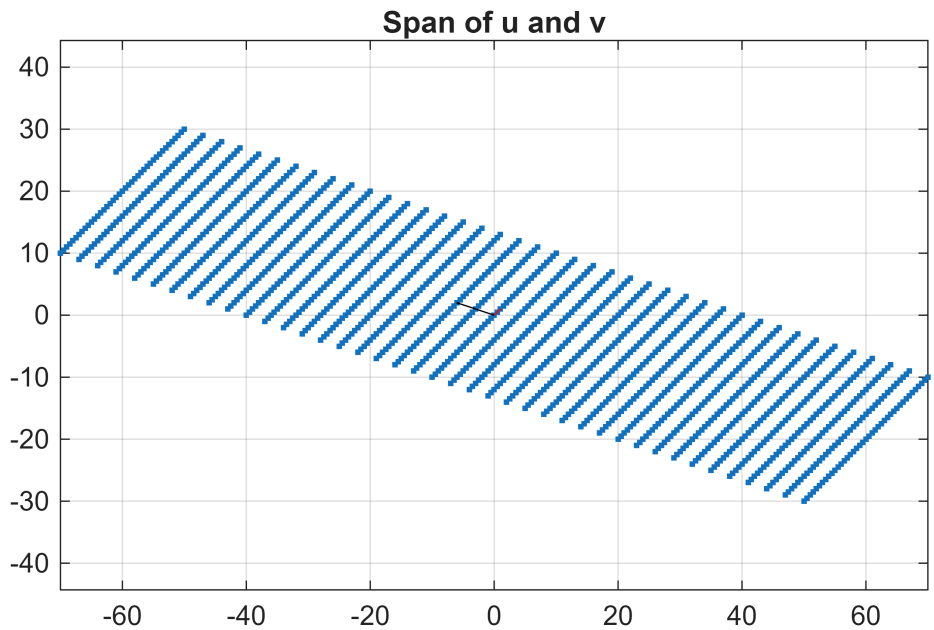
```
clc;
clear;
close all;
u = [1 0];
quiver(0,0,u(1),u(2),0)
grid on; axis equal;
title('u = (1,0)')
```



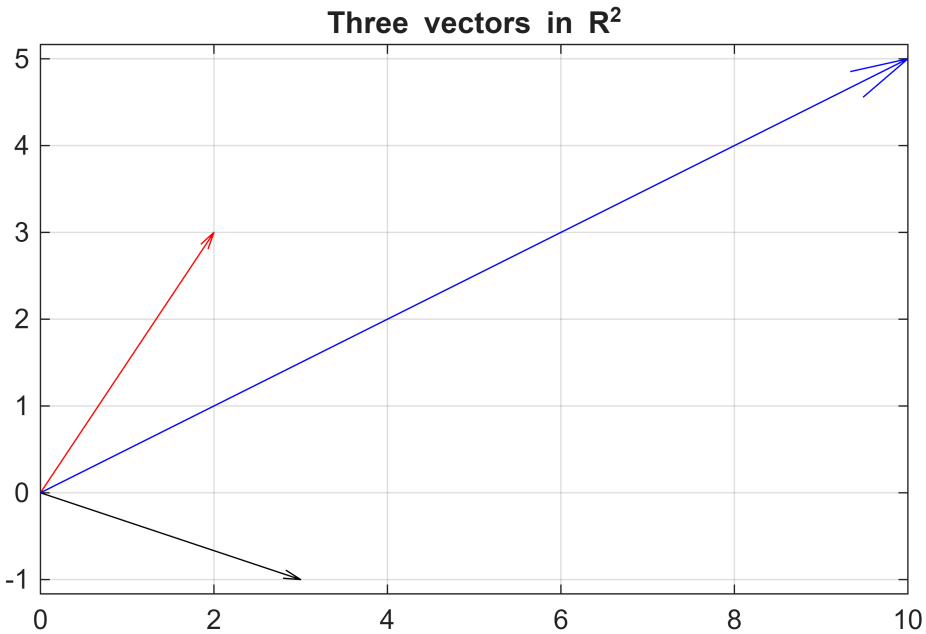
```
clc;
clear;
close all;
u = [12 3];
quiver(0,0,u(1),u(2),0)
grid on; axis equal;
title('u = (12,3)')
```



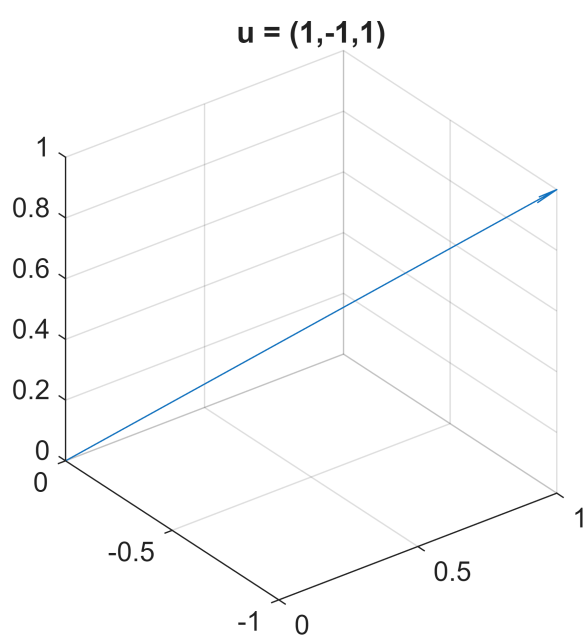

```
clc;
clear;
close all;
u = [1 1];
v = [-6 2];
a = -10:0.5:10;
b = -10:0.5:10;
[k1,k2] = meshgrid(a,b);
w = k1(:)*u + k2(:)*v;
plot(w(:,1),w(:,2),'.')
hold on
quiver(0,0,u(1),u(2),0,'r')
quiver(0,0,v(1),v(2),0,'k')
grid on; axis equal;
title('Span of u and v')
```




```
clc;
clear;
close all;
u1 = [2 3];
u2 = [3 -1];
u3 = [10 5];
quiver(0,0,u1(1),u1(2),0,'r')
hold on
quiver(0,0,u2(1),u2(2),0,'k')
quiver(0,0,u3(1),u3(2),0,'b')
grid on; axis equal;
title('Three vectors in R^2')
```

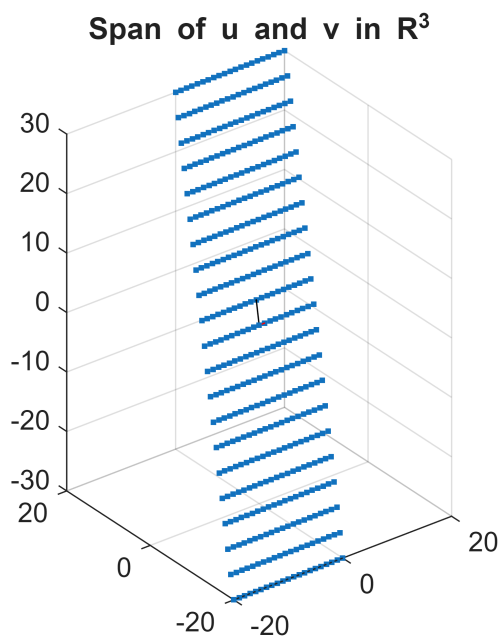



```
clc;
clear;
close all;
u = [1 -1 1];
quiver3(0,0,0,u(1),u(2),u(3),0)
grid on; axis equal; view(3)
title('u = (1,-1,1)')
```




```
clc;
```

```
clear;
close all;
u = [1 0 0];
v = [1 2 3];
a = -10:1:10;
b = -10:1:10;
[k1,k2] = meshgrid(a,b);
w = k1(:)*u + k2(:)*v;
plot3(w(:,1),w(:,2),w(:,3),'.')
hold on
quiver3(0,0,0,u(1),u(2),u(3),0,'r')
quiver3(0,0,0,v(1),v(2),v(3),0,'k')
grid on; axis equal; view(3)
title('Span of u and v in R^3')
```




```
clc;
clear;
close all;
A = [9 12 6
 3 7 5
 7 -10 0];
if rank(A) == 3
 disp('Vectors span R^3 (Independent)')
else
 disp('Vectors are Dependent')
end
```

```
Vectors span R^3 (Independent)
```

```
clc; clear;
```

```
u = [1 2];
v = [1 5];
w = [3 5];
A = [u' v'];
x = A\w';
if norm(A*x - w') < 1e-10
 disp('YES, w is a linear combination')
else
 disp('NO')
end
```

```
YES, w is a linear combination
```

```
%%
clc;
clear;
u1 = [1 1];
u2 = [0 0];
v = [13 -5];
A = [u1' u2'];
x = A\v';
```

```
Warning: Matrix is singular to working precision.
```

```
if norm(A*x - v') < 1e-10
 disp('YES')
else
 disp('NO')
end
```

```
NO
```

```
clc; clear;
u1 = [0 -11 2];
u2 = [21 -5 0];
u3 = [0 0 17];
u = [12 3 -7];
A = [u1' u2' u3'];
x = A\u';
if norm(A*x - u') < 1e-10
 disp('YES')
else
 disp('NO')
end
```

```
YES
```

```
clc;
clear;
u1 = [1 2 -1];
```

```
u2 = [1 3 8];
u3 = [0 1 9];
u = [2 5 7];
A = [u1' u2' u3'];
x = A\u';
```

```
Warning: Matrix is singular to working precision.
```

```
if norm(A*x - u') < 1e-10
 disp('YES')
else
 disp('NO')
end
```

```
NO
```

```
clc;
clear;
A = [2 -3
 1 0];
if rank(A)==2
 disp('Independent')
else
 disp('Dependent')
end
```

```
Independent
```

```
clc;
clear;
A = [2 1 3
 4 5 0];
if rank(A)==2
 disp('Independent')
else
 disp('Dependent')
end
```

```
Independent
```

```
clc;
clear;
A = [1 -3 22
    1 2 0
 2 9 23
 0 0 5];
if rank(A)==3
 disp('Independent')
else
 disp('Dependent')
```

```
end
```

```
Independent
```

```
clc;
clear;
A = [2 3 7
 2 5 9
 2 -1 3];
if rank(A)==3
 disp('Independent')
else
 disp('Dependent')
end
```